Vincent Wang

Robotics Systems | Vision-Language-Action Models | VLA Post-training

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EDUCATION

University of Pennsylvania — **Expected May 2027**

M.S. in Mechanical Engineering and Applied Mechanics - Mechatronics and Robotics

- Coursework: Advanced Dynamics, Mechatronic Systems, Engineering Mathematics, Operating Systems.

East China University of Science and Technology — **June 2025**

B.Eng. in Intelligent Manufacturing Engineering | GPA: 3.6/4.0 | Top 15%

EXPERIENCE

Synthoid.ai | VLA Research Intern — **May 18 - Aug 21, 2026**

Real-robot infrastructure, VLA training, and independent policy-distillation research

- Independently built the robot-side SDK control and VLA adapter for a 7-DoF industrial arm, adding Cartesian end-effector control, multi-rate interpolation, and automatic recovery; delivered stable human teleoperation.
- Led teleoperation and real-robot data collection with a dexterous hand, head/wrist cameras, and LeRobot; synchronized multimodal timestamps and screened abnormal trajectories with first- through third-order action derivatives.
- Resolved abnormal motion from 7-DoF IK ambiguity and traced replay-accuracy failures to an incorrect URDF used by inverse kinematics.
- Deeply contributed to early training of a proprietary VLA model through training implementation/configuration, distributed runs, checkpoint management, initial ablations, and failure diagnosis; identified a gripper-dimension output-contract mismatch and GPU ECC hardware failures.
- Independently defined and implemented Path-OPD for flow-matching VLA distillation, saving the student's inner solver chain and querying frozen-Teacher velocities with gradient isolation. Under matched occupancy, Teacher-query, training, and evaluation budgets, it achieved higher closed-loop success than endpoint DAgger. Preparing an ICLR 2027 submission.

Wuji Tech | Robotics Control Intern — **July - Sept 2024**

Dexterous-hand teleoperation and low-level motor control

- Re-architected an unstable impedance loop into an admittance controller with torque feedback in MATLAB/Simulink for dexterous-hand teleoperation.
- Developed Linux C++ host software for reliable CAN communication and low-level control across approximately 30 motors.

SELECTED PROJECTS, RESEARCH & LEADERSHIP

University of Pennsylvania | Systems & Robotics Projects — **2025 - 2026**

Embedded autonomy and operating-systems engineering

- Built a closed-loop autonomous mobile robot on ESP32/FreeRTOS, integrating sensing, Vive/EKF localization, A* planning, chassis control, mission FSM, and system validation.
- Implemented and debugged PennOS in C, including processes, scheduling, system calls, signals, a file system, and a shell.

ECUST Robocon Team | Co-founder & Control Lead — **Sept 2022 - July 2024**

Autonomous robot systems and interdisciplinary technical leadership

- Built the team's first STM32 motion-control system and integrated Point-LIO, YOLO (approximately 20 FPS embedded), MoveIt, and chassis control into an autonomous robot stack.
- Led a 20-member interdisciplinary team, owned system architecture and technology selection, and directed technical training for 60 new members.

MDATS - Multi-Domain Adaptive Classification for Mechanical Fault Detection — **IEEE MEAE 2024**

- First author; built the PyTorch research pipeline and achieved 88.27% target-domain accuracy, a 40.6-point absolute improvement over baseline.

TECHNICAL SKILLS

Languages & Systems: Python, C/C++, Rust, Linux, Git, CMake, ROS 2, CAN
Robotics & Control: Cartesian control, teleoperation, impedance/admittance control, STM32/ESP32, FreeRTOS, MoveIt
VLA & Training: PyTorch, distributed training, VLA post-training, flow matching, online policy distillation, action-head fine-tuning, RLinf, Pi0.5
Robot Data & Evaluation: LeRobot, multi-camera data, timestamp synchronization, trajectory screening, closed-loop evaluation
Research Operations: Matched experiments, checkpoint management, training diagnostics, CUDA, Teacher-NFE and wall-clock accounting